Pattern files and configuration formats

NOTE Rev A · 2026-10-24

Overview

Every Calib.Tools calibration target includes machine-readable configuration files for common calibration frameworks. This document describes each file format and how to use them.

File formats

Extension	Framework	Description
.opencv.json	OpenCV	JSON with pattern type, dimensions, dictionary
.kalibr.yaml	Kalibr (ETH)	YAML target configuration
.ros2.yaml	ROS 2	Camera calibration parameters
.halcon.descr	MVTec HALCON	Calibration plate description
.dxf	CAD	Board outline for fixture design

OpenCV JSON

The OpenCV JSON file contains all parameters needed for cv2.aruco.CharucoBoard or cv2.findChessboardCorners:

- pattern: "ChArUco" or "Checkerboard"
- rows, cols: grid dimensions
- square_mm, marker_mm: feature sizes in millimetres
- dictionary: ArUco dictionary name
- start_id: first marker ID in this board
- generator: "Calib.Tools"

Kalibr YAML

The Kalibr YAML follows the format expected by kalibr_calibrate_cameras:

- target_type: "aprilgrid" or "checkerboard"
- tagCols, tagRows: grid dimensions
- tagSize, tagSpacing: in metres (SI units)

Custom configurations

Generate custom pattern configurations for any grid size, dictionary, and starting ID using the free pattern generator at calib.tools/tools/generator/.

Coordinate system

All Calib.Tools patterns use the OpenCV coordinate convention:

- Origin: top-left corner of the pattern
- X axis: pointing right
- Y axis: pointing down
- Corner ordering: row-major, matching cv2.findChessboardCorners

For Kalibr, the Y axis is inverted (pointing up). The provided YAML files account for this.

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